

Discovery Executive Summary

Project: disocvery-001 · Generated: 27/08/2026, 13:17:26

Repository: shende-shweta/react-calculator | Branch: main

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The `react-calculator` repository is a minimal frontend-only React 18 SPA: 2 JSX files, 1 presentational component (`App.jsx`, 79 LOC), and no backend layer of any kind. The standard frontend hotspots F1–F5 are all clean — the single component is well within size thresholds, uses modern functional patterns with hooks, has no prop drilling, and makes zero inline API calls. The single significant architectural finding (H10) is that all 7 calculator business operations (`DEL`, `+`, `-`, `/`, `x`, `RESET`, `=`) are completely unimplemented empty stubs inside the component's switch handler, and no custom hook or utility module has been created to own the calculation logic. This leaves the app non-functional: no arithmetic operation produces any result. The dominant risk is **incomplete feature implementation with no separation of concerns** — the component acts as both UI shell and intended-but-absent logic owner, making the codebase structurally fragile for any future contributor who needs to add or test calculator behaviour.

\$1.1 Benchmark Ratings Summary

No backend layer detected — backend hotspots H1–H9 not applicable.

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	79 LOC (1 component)	Good
F2	Missing Frontend Service/Data Layer	Components with inline API/data calls	<10	10–20	>20	0	Good
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0	Good
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2 levels	3–4 levels	>4 levels	0 levels (single component)	Good
F5	Legacy / Inconsistent Component Patterns	Legacy/deprecated-pattern components	0	1–10	>10	0	Good
H10	Missing Calculator Logic Abstraction (additional)	Operations implemented outside component (target: 100% in hook/utility)	100%	50–99%	<50%	0% — 7/7 operations are empty stubs	High Risk

No additional hotspots beyond H10 were observed.

\$1.4 Actions Required

Hotspot	Action	Rating	Priority
H10 Missing Calculator Logic Abstraction	Create <code>src/hooks/useCalculator.js</code> ; implement all 7 operations (<code>DEL</code> , <code>+</code> , <code>-</code> , <code>/</code> , <code>x</code> , <code>RESET</code> , <code>=</code>) with correct state transitions; extract <code>btnValues</code> to <code>src/constants/calculatorButtons.js</code> ; slim <code>App.jsx</code> to consume only <code>{ result, handleInput }</code> from the hook; add <code>useCalculator.test.js</code> covering every operation branch	High Risk	Critical

\$1.5 Expected Outcomes

- **Working calculator:** All 7 operations produce correct state updates once logic is implemented in `useCalculator` , making the app functional end-to-end for its stated purpose.
- **Testable business logic:** The calculation engine can be exercised in isolation via `useCalculator.test.js` without mounting the UI component, enabling fast and deterministic branch coverage.
- **Single-responsibility component:** `App.jsx` becomes a thin view — JSX grid wiring only — so UI changes (themes, layout, keyboard handlers) never touch calculation logic and vice versa.
- **Extensibility without regressions:** New operations (percentage, memory recall, history) are added by extending the hook alone; the component requires no modification.
- **Onboarding clarity:** A contributor landing on the repo immediately sees a `hooks/` boundary and knows exactly where to find and add logic, eliminating the current dead-end empty switch statement.

Report written to `docs/discovery/01-architecture-design.md` . The stack is a **frontend-only React 18 SPA** — H1–H9 backend hotspots are not applicable. F1–F5 are all **Good**. The sole finding is **H10 (High Risk / Critical)**: all 7 calculator operations (`DEL` , `+` , `-` , `/` , `x` , `RESET` , `=`) are empty stubs in `App.jsx` with zero logic implemented anywhere. Fix: extract a `useCalculator` hook, implement the arithmetic, and slim `App.jsx` to a pure view.,"stop_reason":"end_turn","session_id":"68a05741-056b-422f-bdec-b4b8b64e8686","total_cost_usd":0.7326901,"usage":{"input_tokens":18,"cache_creation_input_tokens":41050,"cache_read_input_tokens":781377,"output_tokens":15999,"server_tool_use":{"web_search_requests":0,"web_fetch_requests":0},"service_tier":"standard","cache_creation":{"ephemeral_1h_input_tokens":41050,"ephemeral_5m_input_tokens":0},"inference_geo":"not_available","iterations":[{"input_tokens":1,"output_tokens":1372,"cache_read_input_tokens":62141,"cache_creation_input_tokens":592,"cache_creation":{"ephemeral_5m_input_tokens":0,"ephemeral_1h_input_tokens":592},"type":"message"}],"speed":"standard"},"modelUsage":{"claude-haiku-4-5-20251001":{"inputTokens":11863,"outputTokens":15,"cacheReadInputTokens":0,"cacheCreationInputTokens":0,"webSearchRequests":0,"costUSD":0.011938,"contextWindow":200000,"maxOutputTokens":32000},"cl-sonnet-4-6":{"inputTokens":18,"outputTokens":15999,"cacheReadInputTokens":781377,"cacheCreationInputTokens":41050,"webSearchRequests":0,"costUSD":0.7207521,"contextWindow":200000,"maxOutputTokener[],"terminal_reason":"completed","fast_mode_state":"off","uuid":"0b7a5bf7-f47d-428a-be3b-9dbeb3260b82"}